

WOOHYUK CHOI / Ph.D.

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ABOUT ME

I am a Ph.D. student in Electrical and Computer Engineering at Seoul National University, advised by Prof. Byoungyoung Lee. I designed **Prompt Flow Integrity** and **DualView**, system-level mitigations against prompt injection attacks in LLM agents. I also discovered **Agent Data Injection (ADI)**, a new class of attacks that exploits the misinterpretation of untrusted data as trusted data in AI agents. In addition, I have hands-on experience in system security projects and vulnerability discovery, complementing my main focus on securing modern AI-driven systems.

RESEARCH INTERESTS

I am interested in **security for AI** and **system security** in general. In particular, my research focuses on **AI agent security**, e.g., system-level mitigation against prompt injection attacks for AI agents in several domains.

PUBLICATIONS

- **Agent Data Injection Attacks are Realistic Threats to AI Agents**
Woohyuk Choi*, Juhee Kim*, Taehyun Kang, Jihyeon Jeong, Luyi Xing, and Byoungyoung Lee
arXiv:2607.05120, 2026 (* Equal contribution)
- **DualView: Preventing Indirect Prompt Injection in Personal AI Agents**
Woohyuk Choi*, Juhee Kim*, Taehyun Kang, Youngmin Kim, and Byoungyoung Lee
arXiv:2607.03821, 2026 (* Equal contribution)
- **GHost in the SHELL: A GPU-to-Host Memory Attack and Its Mitigation**
Sihyun Roh, Woohyuk Choi, Jaeyoung Chung, Yoochan Lee, Suhwan Song and Byoungyoung Lee
IEEE Symposium on Security and Privacy (SP) 2026 (accepted, to appear)
- **Prompt Flow Integrity to Prevent Privilege Escalation in LLM Agents**
Woohyuk Choi*, Juhee Kim*, and Byoungyoung Lee
arXiv:2503.15547, 2025 (* Equal contribution)

EXPERIENCE

- Internship, Samsung MX (Mobile eXperience), Security Engineering Group** Jan 2024 - Feb 2024
Samsung MX, Suwon, South Korea
- Developed and tested an eSE-based Digital-Wallet Application for Samsung mobile devices.

PROJECTS

Race Vulnerability Discovery in Linux Kernel Binder System Calls
Platform Stability Committee (Governmental advisory board project)

- Extended Syzkaller with concurrency-aware fuzzing techniques (inspired by Segfuzz, a kernel race fuzzer) to uncover race vulnerabilities in Android Binder syscalls.

Mar 2024 – Aug 2024

EDUCATION

Seoul National University
Seoul, South Korea
Ph.D. in Electrical and Computer Engineering (Advisor: Byoungyoung Lee)

Mar 2025 - Present
Anticipated Feb 2031

Seoul National University
Seoul, South Korea
B.S. in Electrical and Computer Engineering
GPA: 3.95 (Major: 3.99)/4.30 (Summa Cum Laude)

Mar 2019 - Feb 2025
military service (2 years)

AWARDS

Excellence Award, AI-based Vulnerability Discovery System Competition
Korea Institute of Information Security and Cryptology (KIISC), KRW 5,000,000 prize

- Agent CLI Vulnerability Scanner.** Automated tooling to discover security vulnerabilities in AI agent CLI tools by analyzing command validation logic in minified JavaScript code.

Nov 2025

SCHOLARSHIPS

- Electrical and Computer Engineering Foundation (Seoul National University), Full tuition support**

Sep 2025 - Present

TEACHING ASSISTANT

Programming Methodology (430.211)
Seoul National University

Spring 2026

Programming Methodology (430.211)
Seoul National University

Spring 2025

TECHNICAL SKILLS

- Programming Languages: C, C++, Python, JavaScript
- Tools & Frameworks: MCP, LangChain, LangGraph, Syzkaller, AFL, LLVM